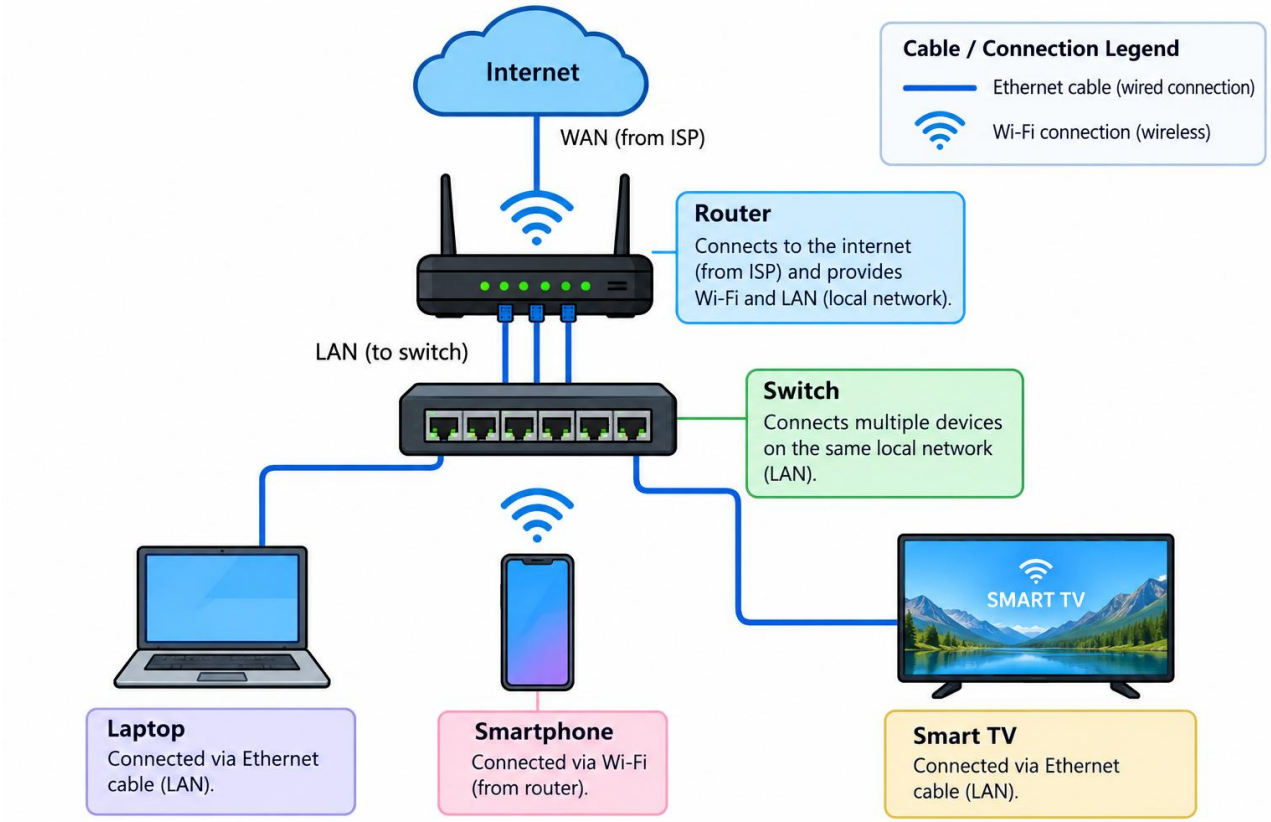
Network Devices Assignment

Q1. Draw and label a simple home network diagram showing a router, switch, and three devices (like a laptop, smartphone, and smart TV), indicating how each device connects to the network.

Answer:



- **Laptop and smartphone:** connect wirelessly through WiFi to the router.
- **Smart TV:** connects with an Ethernet cable to the switch, which is connected to the router.
- **Router:** connects the whole home network to the Internet through the ISP.

Q2. Create a comparison table listing at least four network devices (router, switch, hub, access point) and describe one real-world use case for each, referencing apps or services you use daily (e.g., WiFi for streaming on YouTube, LAN gaming, etc.).

Answer:

Device	What it does	Real-world use case
Router	Connects the home/college network to the Internet and sends data between networks. Most also give IP addresses and have built-in WiFi.	Home WiFi router that lets the family stream YouTube, order on Zomato and use Instagram at the same time.
Switch	Connects wired devices in a LAN and sends data only to the port of the device it is meant for.	LAN gaming at a gaming zone, or college lab computers connected to the lab network, so data goes directly between the right devices.
Hub	Connects wired devices but sends incoming data to all ports. It is simple and slow, and is rarely used now.	Very old small LAN or a basic lab setup for sharing files between a few computers.
Access Point	Creates a WiFi signal for wireless devices and connects them to the wired network.	College campus WiFi in classrooms and the library, so students can use WhatsApp and Instagram on their phones.

Q3. Write a short scenario where you explain how data travels from your phone to the internet when you order food on Zomato, mentioning the role of each network device involved (router, switch, access point).

Answer:

Scenario: I choose my food in the Zomato app and tap "Place Order". My phone sends the order data wirelessly over WiFi to the access point. The access point passes it through an Ethernet cable to the switch, which forwards it to the router. The router sends the data to my ISP, which carries it across the Internet to Zomato's server. The server confirms the order, and the reply travels back along the same path to my phone within a second.

Role of each device:

- **Access Point:** receives the wireless signal from my phone and connects it to the wired network.
- **Switch:** sends the data only to the correct port (towards the router) instead of to every device.
- **Router:** finds the best path and connects my local network to the Internet through the ISP.

Q4. Given this situation: you have a WiFi router, but your gaming PC only has an Ethernet port and is far from the router. Suggest two different network devices you could use to connect your PC to the internet and explain how each would work in this setup.

Answer:

Option 1: Powerline Adapter (pair)

- **How it works:** One adapter is plugged into a power socket near the router and connected to it with an Ethernet cable. The second adapter is plugged in near the gaming PC and connected to the PC with an Ethernet cable. The data travels through the home's electrical wiring between the two adapters.
- **Good point:** No long cable is needed and the connection is more stable than weak WiFi.

Option 2: WiFi Extender with an Ethernet port

- **How it works:** The extender is placed halfway between the router and the PC, where the WiFi signal is still strong. It receives the router's WiFi and passes it to the gaming PC through its Ethernet port using a short Ethernet cable.
- **Good point:** Easy to set up and works without any wiring through the walls.